Last version : 27/10/2026

Finding chaos in randomness

Contents

1. Introduction 1
1.1. Datasets 1
2. Decomposition methods 1
2.1. EMD 1
3. Tools for IMF characterization 2
3.1. runstest 2
3.2. chaostest 3
4. Experiments 3
4.1. First tests 3
4.1.1. Randmoness test 4
4.1.2. Chaos test 5

1. Introduction

[explanations]

1.1. Datasets

- frac_alea : random gaussian white noises, 10 000 points, 10 samples
- ruido_blanco : atmospheric noise, 10 000 points, 10 samples

==> $\boldsymbol{m = 6, \tau = 1}$

2. Decomposition methods

2.1. EMD

[explanations on how it works]

IMF = Intrinsic Mode Function

==> MATLAB implementation with `Signal Multiresolution Analyzer` app and inline function: `[imf,residula,info] = emd(data,varagin{:})`

Parameters:

- SiftRelativeTolerance - Convergence criterion threshold (default: 0.2).
- SiftMaxIterations - Maximum sifting iterations per IMF (default: 100).
- MaxNumIMF - Maximum number of IMFs to extract (default: 10).
- MaxNumExtrema - Extrema count in residual to stop sifting (default: 1).
- MaxEnergyRatio - Maximum signal-to-residual energy ratio (default: 20).
- Interpolation - Envelope interpolation method: “spline” or “pchip”.
- Display - Toggle sifting diagnostic table: 0 or 1 (default: 0).
- SampleRate - Sampling frequency of signal in Hz.

According to the bibliography `Empirical_Mode_Decomposition_-_an_introduction`, we first use emd with `MaxNumIMF = 8` (3 spikes in the residual) and `Interpolation = spline`, others parameters still at default. (empirical quality criteria, to discuss)

3. Tools for IMF characterization

3.1. runstest

Run test for **randomness**. For now, we just use `runstest(data,'Alpha')` with Alpha the significance level. criterias to be discussed

All parameters:

- x (Required Data Input): Vector containing the sequence of data values to be tested for randomness.
- v (Threshold / Cutoff Value): Scalar threshold value used to convert continuous data into a binary sequence of runs (values above vs. values at or below v). If not specified, the sample median of x is used as the default baseline.
- Alpha (Significance Level): Scalar value between 0 and 1 that defines the significance level of the hypothesis test. The default value is 0.05 (5% significance level).
- Method (P-Value Computation Method): String or character vector specifying how the p-value is computed:
  - 'exact' — Computes the exact p-value based on the exact permutation distribution of runs.
  - 'approximate' — Computes the p-value using a normal distribution approximation (recommended for larger datasets).
- Tail (Alternative Hypothesis): String or character vector indicating the direction of the two-sample hypothesis test:

  'both' (Default) — Two-tailed test testing whether the number of runs is significantly higher or lower than expected.

  'right' — One-tailed test testing whether the number of runs is significantly greater than expected.

  'left' — One-tailed test testing whether the number of runs is significantly less than expected.

Example of output: File: IMF_1.mat | h: 1 | p-value: 0.0000 | Runs: 6570 | Decision: REJECT Null (Non-Random)

with:

- h (Hypothesis Decision)

This is a binary indicator representing the final verdict of the statistical test. h = 0: the test fails to reject the null hypothesis. This means there is not enough evidence to say the data is non-random; the sequence behaves like random noise. h = 1: the test

rejects the null hypothesis. This means the data is highly unlikely to be random and contains a distinct pattern, trend, or structured oscillation.

- p-value (Probability Value)

The p-value measures the probability of observing your specific sequence of data assuming the null hypothesis (that the data is perfectly random) is true.A p-value below your chosen significance level (e.g., < 0.05) means the observation is extremely rare for random data, prompting you to reject the assumption of randomness (h = 1).A higher p-value means the data is consistent with randomness.

- Runs

A "run" is an uninterrupted sequence of consecutive data points that all fall on the same side of a cutoff line (typically the sample median). For example, if a signal's values relative to the median are (+, +, -, -, -, +), that represents exactly 3 runs: one group of positives, one group of negatives, and one final positive. A perfectly random signal has a predictable, average number of runs; too many runs indicates high-frequency oscillation, while too few runs indicates a slow trend or flatline.

3.2. chaostest

Run test for **chaos** using Bensaïda MATLAB function. [Explanations of every parameters]

4. Experiments

4.1. First tests

We first generate the EMDs with standard parameters `EMD_gen(frac_alea(1,:),'database/frac_alea/frac_alea_EMD_1','MaxNumIMF',8);`
Then we plot the decomposition of the first frac_alea signal on the HxC plane, leading to:

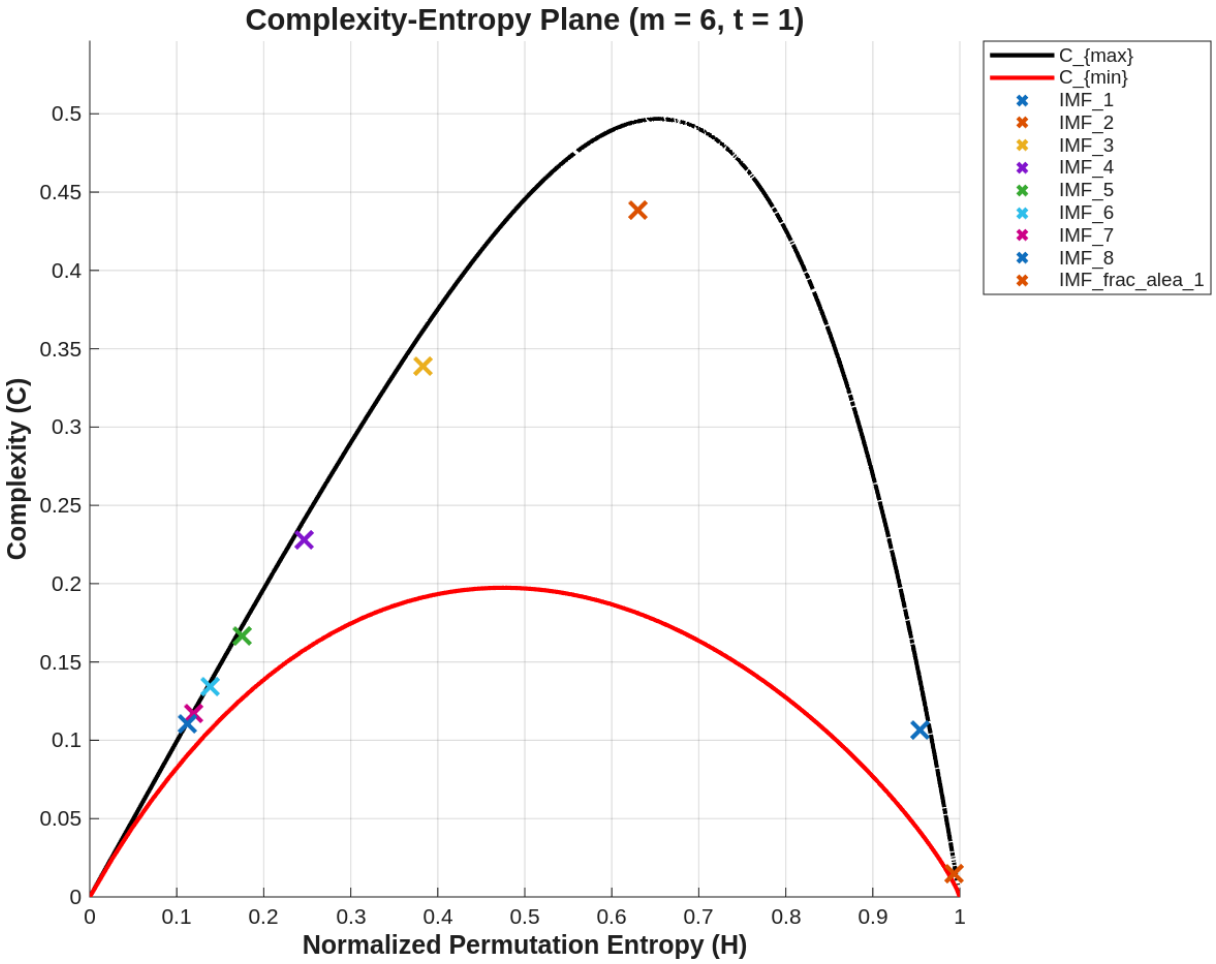


Figure 1: IMFs positionning on the Entropy-Complexity plane from the decomposition of frac_alea_1

RQ: IMF_frac_alea_1 is the original signal.

Observations: 3 types of IMF: random (1), chaotic (2 and 3 maybe), deterministic (4 ->8)

Next step: Categorize each IMF (random/chaotic/deterministic) and check the labelling with `runstest`, `chaostest` and maybe Lyapunov Exponent (which F_s?)

4.1.1. Randmoness test

For now, `alpha =0.05`

File	h	p-value	runs	Decision
IMF_1	1	0	6570	Non-Random
IMF_2	1	0	2292	Non-Random
IMF_3	1	0	817	Non-Random
IMF_4	1	0	277	Non-Random
IMF_5	1	0	68	Non-Random
IMF_6	1	0	11	Non-Random
IMF_7	0	1	1	Random
IMF_8	0	1	1	Random

Table 1: Randomness classification for the EMD of frac_alea_1

The test characterizes only IMF_7 and IMF_8 as random, whereas IMF 4,5 and 6 seem to be good candidates for being random. To be corrected

4.1.2. Chaos test

For now we just tried with default parameters ie:

- ActiveFN:'tanh' Activation function used in the neural network estimation ('tanh', 'logistic', or 'sigmoid')
- maxOrders[5, 6, 5] Vector specifying maximum model orders $[L, m, q]$.
- alpha: 0.05 Significance level for the test (5%).
- flag: 'VARY' Mode specifying whether to vary order triplets $[L, m, q]$ ('VARY') or fix them ('FIX').To be discussed

Nb : the lyapunov exponent is computed without the use of the associated matlab function => no sampling frequency

We obtain

File	h	p-value	λ	Orders	Decision
IMF_1	1	0	-0.2858	[1, 6, 4]	Non-Chaotic
IMF_2	1	0	-0.0800	[1, 3, 1]	Non-Chaotic
IMF_3	0	0.2123	-0.0102	[1, 3, 2]	Chaotic
IMF_4	0	1	0.0401	[1, 4, 1]	Chaotic
IMF_5	0	1	0.6365	[2, 3, 1]	Chaotic
IMF_6	0	1	0.7834	[3, 5, 1]	Chaotic
IMF_7	0	1	0.7817	[2, 3, 1]	Chaotic
IMF_8	0	1	0.7834	[5, 3, 1]	Chaotic

Table 2: Chaos classification for the EMD of frac_alea_1

Observations: Efficient for rejecting randomness but not deterministic behavior. Need to tune it